In this page, we follow the book of lars-ahlfors.

Def. Let an arc γ be given by equation $Z = Z(t)$ $\alpha \leq t \leq \beta$ in a region Ω.
Let f be a continuous map which values defined in Ω.
An arc γ' which is given by $W = f(Z(t))$ is called the image of γ

Consider the f is analytic and $Z(t)$ is smooth. Then,

$$W'(t) = f'(Z(t)) \cdot Z'(t)$$

For t_0, if $Z'(t_0) \neq 0$ and $f'(Z(t_0)) \neq 0$, then $W'(t_0) \neq 0$. Now,

$$\arg W'(t_0) = \arg f'(Z_0) + \arg Z'(t_0).$$

This equation means the difference of the tangent angle of the curve γ between the image γ' is exactly $\arg f'(Z_0)$, regardless of the curve.

「Thm.

Suppose that $f(z)$ is analytic at z_0 with $f'(z_0) \neq 0$.
Let $C_1 : z_1(t)$ and $C_2 : z_2(t)$ be smooth curves in the domain
that intersect at $z_0 = z_1(t_0) = z_2(t_0)$, with $C_1' : w_1(t)$ and $C_2' : w_2(t)$,
the images of C_1 and C_2, respectively.
Then, the angle between C_1 and C_2 measured from C_1 to C_2 is equal to
the angle between C_1' and C_2' measured from C_1' to C_2'.